

1.1. Technical character of an invention

It has been part of the European legal tradition that patent protection should be reserved for technical creations. It was indeed always common ground that creations in engineering and technology were entitled to patent protection under the EPC (see e. g. T 22/85, OJ 1990, 12; T 154/04, OJ 2008, 46). The use of the term "invention" in Art. 52(1) EPC 1973 in conjunction with the so-called "exclusion provisions" of Art. 52(2) and (3) EPC 1973 was interpreted by the boards of appeal as **implying** a requirement of technical character or technicality to be fulfilled by an invention as claimed in order to be patentable (T 931/95, T 1173/97 and T 935/97). Thus an invention may be an invention within the meaning of Art. 52(1) EPC if, for example, a technical effect is achieved by the invention or if **technical considerations** are required to carry out the invention. In T 1173/97 (OJ 1999, 609) and T 935/97 the board stated that within the context of the application of the EPC the technical character of an invention was generally accepted as an essential requirement for its patentability. In T 931/95 (OJ 2001, 441) the board held that technical character was an implicit requirement of the EPC to be met by an invention in order to be an invention within the meaning of Art. 52(1) EPC 1973 (see also T 1543/06).

In T 154/04 (OJ 2008, 46) the board summarised the main principles underlying the relevant jurisprudence of the boards of appeal. The board held that "technical character" was an **implicit requisite** of an "invention" within the meaning of Art. 52(1) EPC 1973 (requirement of "technicality"). Art. 52(2) EPC 1973 did not exclude from patentability any subject matter or activity having a technical character, even if linked to items listed in this provision since these items were **only excluded "as such"** (Art. 52(3) EPC 1973). In examining the patentability of an invention in respect of a claim, the claim had to be construed to determine the **technical features** of the invention, i.e. the features which contributed to the technical character of the invention (see also T 931/95, OJ 2001, 441, T 914/02 and T 1543/06). The board observed that the intention of Art. 52(3) EPC 1973 was clearly to ensure that anything which had previously been a patentable invention under conventional patentability criteria should remain patentable under the EPC.

In T 619/02 (OJ 2007, 63) the board stated that, as had long been accepted, (see in this respect decisions T 22/85, OJ 1990, 12, T 931/95, OJ 2001, 441, and T 258/03, OJ 2004, 575), the technical character of an invention constituted a fundamental prerequisite inherent to Art. 52 EPC 1973 – and was formulated as an explicit requirement in Art. 52(1) EPC. Thus, technical character constituted a **conditio sine-qua-non** to be met by an invention susceptible of patent protection under the EPC. Additionally, the board noted that a technical invention is generally useful and practical within the generic meaning of these terms, but the reverse is not necessarily true, i.e. not every "practical art" and not every entity or activity that is practical or useful is necessarily technical in the patent law sense (see T 388/04, OJ 2007, 16). It was noted that in some national patent law systems, "useful" is the counterpart to the requirement of industrial applicability set out in Art. 57 EPC 1973 (see Note 5 to Art. 27(1) TRIPS); however, this special meaning of the term does not necessarily convey technical character either. Accordingly, considerations of usefulness and practicality are no substitute for, or criteria equivalent to the prerequisite of technical character inherent in the EPC. The board observed that although industrial

applicability and technical character of an invention are intimately related to each other (T 854/90), they are not synonymous and constitute two distinct, non-equivalent requirements under the EPC 1973 (T 953/94).

1.2. Separate and independently patentability requirements

Art. 52(1) EPC 1973 in conjunction with Art. 52(2) and (3) EPC 1973 is generally interpreted as implying a requirement of technical character for a claimed invention to be patentable. Whereas technical character was previously assessed using the "contribution approach", recent case law has abandoned this approach in favour of one which recognises the requirement of technical character as separate and independent of the remaining requirements of Art. 52(1) EPC, in particular novelty and inventive step, and compliance with which can therefore be assessed without having recourse to the prior art (T 1543/06).

According to T 154/04 (OJ 2008, 46), it was clear from the wording of Art. 52(1) EPC 1973 and the use of the term "invention" in the context of the criteria for patentability, that the requirements of invention, novelty, inventive step, and susceptibility of industrial application were separate and independent criteria, which might give rise to concurrent objections under any of these requirements. Novelty, in particular, was not a requisite of an invention within the meaning of Art. 52(1) EPC 1973, but a separate patentability requirement. This construction of Art. 52(1) EPC 1973 had a clear basis in the case law of the Enlarged Board of Appeal (G 2/88, OJ 1990, 93 and G 1/95, OJ 1996, 615).

1.3. Absolute and relative patentability requirements

In T 154/04, OJ 2008, 46, the board indicated that the examination of whether there was an invention within the meaning of Art. 52(1) to (3) EPC 1973 should be strictly separated from and not confused with the examination of the other three patentability requirements referred to in Art. 52(1) EPC 1973. It distinguished the concept of "invention" as a general and absolute requirement of patentability from the relative criteria, novelty and inventive step, (which, in an ordinary popular sense, were understood to be the attributes of any invention), and the requirement of industrial applicability. Decisive for the presence of a (potentially patentable) invention was the inherent character of the claimed subject-matter.

1.4. Verification of the presence of an invention under Article 52(1) EPC

In T 258/03, OJ 2004, 575 (see also T 154/04, OJ 2008, 46), the board stated that the verification that claimed subject-matter is an invention within the meaning of Art. 52(1) EPC 1973 is, in principle, a **prerequisite** for the examination in respect of novelty, inventive step and industrial application, since these latter requirements were defined only for inventions (see Art. 54(1), 56, and 57 EPC 1973). The structure of the EPC therefore suggests that it should be possible to determine whether subject-matter was excluded under Art. 52(2) EPC 1973 without any knowledge of the state of the art (including common general knowledge).

1.4.1 Assessment of the invention independently of the prior art

a) Earlier case law of the boards of appeal: the "contribution approach"

In order to assess whether the claimed subject-matter was an invention within the meaning of Art. 52(1) EPC 1973, the boards of appeal applied, in their earlier case law, the so-called "contribution approach", according to which an invention had a technical character if it provided a technical contribution to the state of the art in a field not excluded from patentability under Art. 52(2) EPC 1973 (see T 121/85, T 38/86, OJ 1990, 384; T 95/86, T 603/89, OJ 1992, 230; T 71/91, T 236/91, T 833/91, T 77/92). The idea behind the so-called contribution approach was that the EPC 1973 only permitted patenting "in those cases in which the invention involves some contribution to the art in a field not excluded from patentability" (T 38/86, OJ 1990, 384). In other words, in order to assess the first requirement, i.e. the presence of an invention within the meaning of Art. 52(1) EPC 1973, a criterion was established which relied on meeting further requirements mentioned in that Article, in particular novelty and/or inventive step. Thus, some prior art was taken into account when determining whether subject-matter was excluded under Art. 52(2) and (3) EPC 1973 (T 258/03, OJ 2004, 575; see T 769/92, OJ 1995, 525).

b) Abandonment of the "contribution approach"

Subsequent decisions found any comparison with the prior art to be inappropriate for determining whether or not an invention was present.

In T 1173/97 (OJ 1999, 609) the board consciously abandoned the "contribution approach" and found that compliance with Art. 52(2), (3) EPC can be established by features already known in the art. Determining the technical contribution an invention achieves with respect to the prior art is more appropriate for the purpose of examining novelty and inventive step than for deciding on possible exclusion under Art. 52(2) and (3) EPC 1973. In T 931/95 (OJ 2001, 441) it was stated that there is no basis in the EPC for distinguishing between "new features" of an invention and features of that invention which are known from the prior art, when examining whether the invention concerned may be considered to be an invention within the meaning of Art. 52(1) EPC 1973. Thus there is no basis in the EPC for applying the so-called contribution approach for this purpose (see also; T 1001/99, T 388/04, OJ 2007, 16; T 619/02, OJ 2007, 63; T 553/02; T 258/03, OJ 2004, 575).

c) Some consequences of abandoning the contribution approach

Over a series of decisions the boards of appeal (and in particular Board 3.5.01) explored the consequences of abandoning the contribution approach (see G 3/08, OJ 2011, 10).

In T 931/95 (OJ 2001, 441), the board decided that an apparatus for carrying out an activity excluded as such from patentability by Art. 52(2) and (3) EPC 1973 was not itself excluded from patentability. In particular, a claim directed to a computer loaded with a program was not excluded from patentability by Art. 52(2) EPC 1973 even if the program itself would be, i.e. if the program caused no "further technical effect" when run. That

decision did not however extend the logic to methods employing technical means (Headnote 2). The board also held that, for the purpose of determining the extent of the exclusion under Art. 52(2) and (3) EPC 1973, the "further" technical effect could, in its opinion, be known in the prior art (see also T 1461/12, T 556/14).

With regard to methods, this decision was explicitly overturned by T 258/03 (Headnote 1); T 258/03 came to the conclusion that any claim involving technical means was not excluded from patentability by Art. 52(2) EPC 1973 (see points 3 and 4 of the Reasons). T 424/03, finally extended the reasoning applied in T 258/03 to come to the conclusion that a claim to a program ("computer executable instructions" in the claim in question) on a computer-readable medium also necessarily avoids exclusion from patentability under Art. 52(2) EPC 1973 (see Catchword 2 and point 5.3 of the Reasons).

1.4.2 Technical considerations and technical implementations

There might be an invention within the meaning of Art. 52(1) EPC 1973 if, for example, a **technical effect** was achieved by the invention or if **technical considerations** were required to carry out the invention (see T 931/95, OJ 2001, 441).

In G 3/08 (OJ 2011, 10) the Enlarged Board of Appeal could not identify a divergence in the case law of the boards of appeal, but noted that there was at least the potential for confusion, arising from the assumption that **any technical considerations** are sufficient to confer technical character on claimed subject-matter, a position which was apparently adopted in some cases (e.g. T 769/92).

In T 1173/97 the board noted that all computer programs have technical effects, since for example when different programs are executed they cause different electrical currents to circulate in the computer they run on. However such technical effects are not sufficient to confer "technical character" on the programs; they must cause **further technical effects**.

Thus, although it may be said that all computer programming involves technical considerations since it is concerned with defining a method which can be carried out by a machine, that in itself is not enough to demonstrate that the program which results from the programming has technical character; the programmer must have had technical considerations beyond "merely" finding a computer algorithm to carry out some procedure (G 3/08, point 13.5 of the Reasons; see also T 1358/09).

In T 914/02 the involvement of technical considerations, however, was **not sufficient** for a method, which might exclusively be carried out mentally, to have technical character. Technical character could be provided through the technical implementation of the method, resulting in the method providing a **tangible, technical effect**, such as the provision of a physical entity as the resulting product, or a non-abstract activity, such as through the use of technical means. The board rejected a claim directed to an invention involving technical considerations and encompassing technical embodiments on the grounds that the invention as claimed could also be exclusively performed by purely mental acts excluded from patentability under Art. 52(2)(c) EPC 1973.

In T.619/02 (OJ 2007, 63) the board stated that the prerequisite of technical character inherent to the EPC cannot be considered to be fulfilled by an invention, as claimed, which, although possibly encompassing technical embodiments, also encompasses ways of implementing it that do not qualify as technical. It followed that an invention is susceptible of patent protection within the meaning of Art. 52(1) EPC 1973 if, and only if the invention as claimed includes aspects that impart technical character to essentially all ways of performing the invention. An analogous condition applies in fact to the exclusions from patentability under Art. 52(2) EPC 1973, as illustrated by decision T.914/02 in which the corresponding board rejected a claim directed to an invention involving technical considerations and encompassing technical embodiments on the grounds that the invention as claimed could also be exclusively performed by purely mental acts excluded from patentability under Art. 52(2)(c) EPC (see also T.388/04, OJ 2007, 16, point 3 of the Reasons).

In T.306/04 the board pointed out that **the mere possibility** of serving a technical purpose or of solving a technical problem was not sufficient to avoid exclusion under Art. 52(2) and (3) EPC 1973 (see also T.388/04, OJ 2007, 16; T.1410/07).

In T.471/05 the fact that the claimed method encompassed non-excluded implementations did not overcome the fact that the claimed method also encompassed excluded subject-matter (T.453/91, T.914/02, points 2 and 3 of the Reasons; T.388/04, OJ 2007, 16; T.930/05). Thus, as long as the claimed design method was not confined to physical, technical implementations, the claimed subject-matter encompassed embodiments excluded from patentability under Art. 52(1) to 52(3) EPC 1973 and was not entitled to patent protection under the EPC.

1.4.3 Methods involving technical means

It is established case law of the boards of appeal that claimed subject-matter specifying at least one feature not falling within the ambit of Art. 52(2) EPC is not excluded from patentability by the provisions of Art. 52(2) and (3) EPC (G 3/08, OJ 2011, 10; T.258/03, T.424/03, T.1658/06, T.313/10).

According to T.258/03 (OJ 2004, 575) technical character results either from the physical features of an entity or (for a method) from the use of technical means. T.258/03 came to the conclusion that any claim involving technical means is not excluded from patentability by Art. 52(2) EPC, and since a claim directed to a method of operating a computer involved a computer it could not be excluded from patentability by Art. 52(2) EPC (see also G 3/08). The board noted that previous decisions of the boards of appeal had held that the use of technical means for carrying out a method for performing mental acts, partly or entirely without human intervention, might, having regard to Art. 52(3) EPC 1973, render such a method a technical process or method and therefore an invention within the meaning of Art. 52(1) EPC 1973 (T.38/86, OJ 1990, 384; T.769/92). However, method steps consisting of modifications to a business scheme and aimed at circumventing a technical problem rather than solving it by technical means could not contribute to the technical character of the subject-matter claimed.